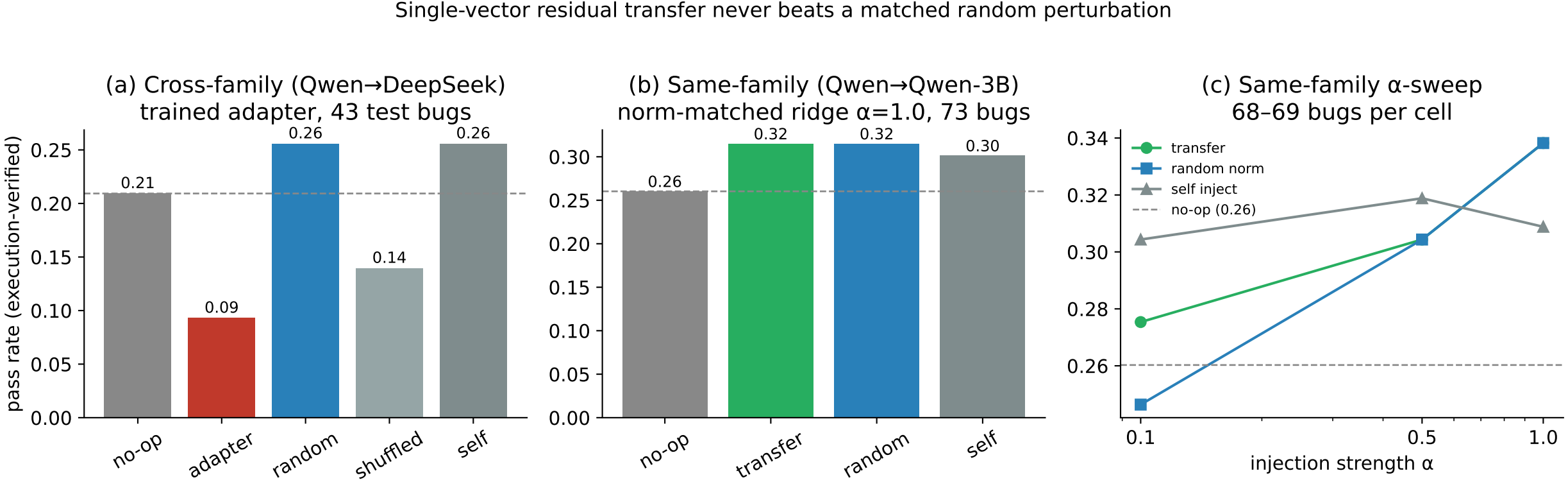
Single-vector residual transfer never beats a matched random perturbation
(a) Cross-family (Qwen→DeepSeek)
trained adapter, 43 test bugs
pass rate (execution-verified)
0.00
0.05
0.10
0.15
0.20
0.25
0.21
0.09
0.26
0.14
0.26
no-op
adapter
random
shuffled
self
(b) Same-family (Qwen→Qwen-3B)
norm-matched ridge α=1.0, 73 bugs
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.26
0.32
0.32
0.30
no-op
transfer
random
self
(c) Same-family α-sweep
68–69 bugs per cell
0.26
0.28
0.30
0.32
0.34
transfer
random norm
self inject
no-op (0.26)
0.1
0.5
1.0
injection strength α